



# Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)  
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

## Test Report

Sample Number: VEL/HCL/A/07  
Name & Address of the Party: M/s Hindustan Copper Limited,  
Indian Copper Complex,  
Ghatsila, Disst.- Singhbhum,  
Jharkhand India.

Report No.: VEL/A/2106/24/007  
Format No.: 7.8 F-01  
Party Reference No.: NIL  
Reporting Date: 29/06/2021  
Period of Analysis: 24/06/2021 to 29/06/2021  
Receipt Date: 24/06/2021

Sample Description: AMBIENT AIR QUALITY MONITORING

### General Information:-

Sample collected by : Vardan EnviroLab Representative (Mr. Kamal)  
Sampling Location : Kendadih Mines Near Sub Station  
Instrument Used : RDS & FPS Sampler with all Accessories  
Instrument Code : VEL/RDS/10 & VEL/FPS/10  
Instrument Calibration Status : Calibrated  
Meteorological condition during monitoring : Clear Sky  
Date of Monitoring : 21/06/2021 to 22/06/2021  
Time of Monitoring : 01:05 PM – 01:05 PM  
Ambient Temperature (°C) : Min.21°C & Max. 33°C  
Surrounding Activity : Human, Vehicular & Construction Activities  
Sampling & Analysis Protocol : IS:5182 & CPCB Guidelines  
Parameter Required : As per Work Order

| S. No. | Parameter                               | Protocol                                | Result | Unit              | NAAQS*<br>Limit |
|--------|-----------------------------------------|-----------------------------------------|--------|-------------------|-----------------|
| 1.     | Particulate Matter (PM <sub>2.5</sub> ) | #SOP No. VEL/SOP/01, Section No. SP 63  | 38.52  | µg/m <sup>3</sup> | 60              |
| 2.     | Particulate Matter (PM <sub>10</sub> )  | IS: 5182 (P-23), Gravimetric Method     | 72.35  | µg/m <sup>3</sup> | 100             |
| 3.     | Nitrogen Dioxide (NO <sub>x</sub> )     | IS: 5182 (P-6), Jacob & Hochheiser      | 24.05  | µg/m <sup>3</sup> | 80              |
| 4.     | Sulphur Dioxide (SO <sub>2</sub> )      | IS: 5182 (P-2), Modified West and Gaeke | 9.08   | µg/m <sup>3</sup> | 80              |

\* NAAQS – National Ambient Air Quality Standards: Schedule-VII, [Rule 3 (3B)], [Part-II-sec.-3(i)] 18.11.2009. #SOP- Laboratory Standard Operating Procedure  
Remark:-All parameters are under limits as per CPCB NAAQS 2009. Instruments Used – Respirable Dust Sampler (APM 460), Calibration Date- 17/05/21,  
Fine Particulate Sampler (EI 133), Calibration Date- 17/05/21

KOMAL SINGH  
ANALYST  
(Tested By)

ARJUN RAWAT  
(Checked By)

Gaurav  
Pratap Singh  
(Approved By)  
Authorized Signatory



www.vardan.co.in

Ph: 0124-4343750/752/753, 9810355569, 9953147268 E-mail: lab@vardanenvironet.com, bd@vardanenvironet.com



# Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)  
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

## Test Report

Sample Number: VEL/HCL/A/08  
Name & Address of the Party: M/s Hindustan Copper Limited,  
Indian Copper Complex,  
Ghatsila, Disst.- Singhbhum,  
Jharkhand India.

Report No.: VEL/A/2106/24/008  
Format No.: 7.8 F-01  
Party Reference No.: NIL  
Reporting Date: 29/06/2021  
Period of Analysis: 24/06/2021 to 29/06/2021  
Receipt Date: 24/06/2021

Sample Description: FUGITIVE AIR QUALITY MONITORING

### General Information:-

Sample collected by : Vardan EnviroLab Representative (Mr. Kamal)  
Sampling Location : Kendadih Mines Under Ground 5 Level  
Instrument Used : RDS & FPS Sampler with all Accessories  
Instrument Code : VEL/RDS/09 & VEL/FPS/09  
Instrument Calibration Status : Calibrated  
Meteorological condition during monitoring : Clear Sky  
Date of Monitoring : 22/06/2021  
Time of Monitoring : 02:21 PM – 10:21 PM  
Ambient Temperature (°C) : Min.25°C & Max. 32°C  
Surrounding Activity : Human, Vehicular & Construction Activities  
Sampling & Analysis Protocol : IS:5182 & CPCB Guidelines  
Parameter Required : As per Work Order

| S. No. | Parameter                               | Protocol                                | Result | Unit              | Limits<br>As per factory Act 1948,<br>Schedule-II<br>(Sec-41F) |
|--------|-----------------------------------------|-----------------------------------------|--------|-------------------|----------------------------------------------------------------|
| 1.     | Particulate Matter (PM <sub>2.5</sub> ) | #SOP No. VEL/SOP/01, Section No. SP 63  | 0.063  | mg/m <sup>3</sup> | --                                                             |
| 2.     | Particulate Matter (PM <sub>10</sub> )  | IS: 5182 (P-23), Gravimetric Method     | 0.086  | mg/m <sup>3</sup> | --                                                             |
| 3.     | Nitrogen Dioxide (NO <sub>x</sub> )     | IS: 5182 (P-6), Jacob & Hochheiser      | 0.032  | mg/m <sup>3</sup> | 6                                                              |
| 4.     | Sulphur Dioxide (SO <sub>2</sub> )      | IS: 5182 (P-2), Modified West and Gaeke | 0.025  | mg/m <sup>3</sup> | 5                                                              |

Remark:-All parameters are under limits as per Factory Act 1948, Schedule-II (Sec 41 F). Instruments used – Respirable Dust Sampler (APM460),  
Calibration Date-17/05/21, Fine Particulate Sampler (EI 133), Calibration Date- 17/05/21

KOMAL SINGH  
(Checked By)

ARJUN RAWAT  
(Checked By)

Gaurav  
Patel Singh  
(Approved By)  
Authorized Signatory



www.vardan.co.in



# Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)  
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

## Test Report

Sample Number: VEL/HCL/A/09  
Name & Address of the Party: M/s Hindustan Copper Limited,  
Indian Copper Complex,  
Ghatsila, Disst.- Singhbhum,  
Jharkhand India.

Report No.: VEL/A/2106/24/009  
Format No.: 7.8 F-01  
Party Reference No.: NIL  
Reporting Date: 29/06/2021  
Period of Analysis: 24/06/2021 to 29/06/2021  
Receipt Date: 24/06/2021

Sample Description: FUGITIVE AIR QUALITY MONITORING

### General Information:-

Sample collected by : Vardan EnviroLab Representative (Mr. Kamal)  
Sampling Location : Kendadih Mines Near Dumping Yard  
Instrument Used : RDS & FPS Sampler with all Accessories  
Instrument Code : VEL/RDS/10 & VEL/FPS/10  
Instrument Calibration Status : Calibrated  
Meteorological condition during monitoring : Clear Sky  
Date of Monitoring : 22/06/2021  
Time of Monitoring : 01:55 PM – 09:55 PM  
Ambient Temperature (°C) : Min.25°C & Max. 32°C  
Surrounding Activity : Human, Vehicular & Construction Activities  
Sampling & Analysis Protocol : IS:5182 & CPCB Guidelines  
Parameter Required : As per Work Order

| S. No. | Parameter                               | Protocol                                | Result | Unit              | Limits<br>As per factory Act 1948,<br>Schedule-II<br>(Sec-41F) |
|--------|-----------------------------------------|-----------------------------------------|--------|-------------------|----------------------------------------------------------------|
| 1.     | Particulate Matter (PM <sub>2.5</sub> ) | SOP No. VEL/SOP/01, Section No. SP 63   | 0.057  | mg/m <sup>3</sup> | --                                                             |
| 2.     | Particulate Matter (PM <sub>10</sub> )  | IS: 5182 (P-23), Gravimetric Method     | 0.091  | mg/m <sup>3</sup> | --                                                             |
| 3.     | Nitrogen Dioxide (NO <sub>x</sub> )     | IS: 5182 (P-6), Jacob & Hochheiser      | 0.036  | mg/m <sup>3</sup> | 6                                                              |
| 4.     | Sulphur Dioxide (SO <sub>2</sub> )      | IS: 5182 (P-2), Modified West and Gacke | 0.022  | mg/m <sup>3</sup> | 5                                                              |

Remark:-All parameters are under limits as per Factory Act 1948, Schedule-II (Sec 41 F). Instruments used – Respirable Dust Sampler (APM460),  
Calibration Date-17/05/21, Fine Particulate Sampler (EI 133), Calibration Date- 17/05/21

KOMAL SINGH  
ANALYST  
(Tested By)

ARJUN RAWAT  
(Checked By)



www.vardan.co.in

Ph: 0124-4343750/752/753, 9810355569, 9953147268 E-mail: lab@vardanenvironet.com, bd@vardanenvironet.com



# Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)  
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

## Test Report

Sample Number: VEL/HCL/AN/05  
Name & Address of the Party: M/s Hindustan Copper Limited,  
Indian Copper Complex,  
Ghatsila, Disst.- Singhbhum,  
Jharkhand India.

Report No.: VEL/AN/2106/24/005  
Format No.: 7.8 F-01  
Party Reference No.: NIL  
Reporting Date: 29/06/2021  
Receipt Date: 24/06/2021

Sample Description: AMBIENT NOISE LEVEL MONITORING

### General Information:-

Sample collected by : Vardan EnviroLab Representative (Mr. Kamal)  
Sampling Location : Kendadih Mines Near Sub Station  
Instrument Used : Sound Level Meter  
Instrument Calibration Status : Calibrated  
Meteorological condition during monitoring : Clear Sky  
Date of Monitoring : 21/06/2021 to 22/06/2021  
Time of Monitoring : 06:00 AM to 06:00 AM  
Ambient Temperature (°C) : Min.21°C & Max. 33°C  
Surrounding Activity : Human, Vehicular, Plant & Other Activities  
Sampling & Analysis Protocol : IS:9989  
Sampling Duration : 24 Hours  
Parameter Required : As per Work Order

| S. No. | Parameters                                     | Protocol | Test Result dB (A)                 |                                      | Unit   |
|--------|------------------------------------------------|----------|------------------------------------|--------------------------------------|--------|
|        |                                                |          | Day Time<br>(06:00 am to 10:00 pm) | Night Time<br>(10:00 pm to 06:00 am) |        |
| 1.     | $L_{max}$                                      | IS 9989  | 74.6                               | 69.8                                 | dB (A) |
| 2.     | $L_{min}$                                      | IS 9989  | 56.8                               | 49.3                                 | dB (A) |
| 3.     | $L_{eq}$                                       | IS 9989  | 64.92                              | 59.82                                | dB (A) |
| 4.     | CPCB Limits in dB(A*) Leq<br>(Industrial Area) | -        | 75.00                              | 70.00                                | dB(A)  |

Note: \*A "decibel" is a unit in which noise is measured.

Remark: All parameters are under limits as per CPCB Noise Standards 2000.  
Instruments Used - Noise Meter (Q657612), Calibration Date- 17/05/21.

KOMAL SINGH  
ANALYST  
(Tested By)

ARJUN RAWAT  
(Checked By)



www.vardan.co.in

Ph: 0124-4343750/752/753, 9810355569, 9953147268 E-mail: lab@vardanenvironet.com, bd@vardanenvironet.com



# Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)  
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

## Test Report

**Sample Number:** VEL/HCL/W/04  
**Name & Address of the Party:** M/s Hindustan Copper Limited,  
Indian Copper Complex,  
Ghatsila, Disst.- Singhbhum,  
Jharkhand India.  
**Report No.:** VEL/W/2106/24/004  
**Format No.:** 7.8 F-01  
**Party Reference No.:** NIL  
**Reporting Date:** 29/06/2021  
**Period of Analysis:** 24/06/2021 to 29/06/2021  
**Receipt Date:** 24/06/2021  
**Sample Description:** Ground Water Sample  
**Sampling Location:** Borewell Water (Kendadih Mines)  
**Sampling Date:** 21/06/2021  
**Sample Collected By:** VEL Representative (Mr. Kamal)  
**Sampling Quantity:** Grab  
**Sampling & Analysis Protocol:** IS 3025 & APHA, 23<sup>rd</sup> Edition 2017  
**Sampling Type:** 2.0 Ltr. + 200 ml  
**Parameter Required:** As Per Work Order  
**Preservation:** Refrigerated

| S. No. | Parameter                           | Test Method                                      | Result                | Unit  | Limits of IS:10500 -2012             |                                                                  |
|--------|-------------------------------------|--------------------------------------------------|-----------------------|-------|--------------------------------------|------------------------------------------------------------------|
|        |                                     |                                                  |                       |       | Requirement<br>(Acceptable<br>Limit) | Permissible<br>limit in the<br>Absence of<br>Alternate<br>Source |
| 1.     | pH (at 25°C)                        | APHA, 4500 H <sup>+</sup> B Electrometric Method | 7.63                  | --    | 6.5 to 8.5                           | No Relaxation                                                    |
| 2.     | Colour                              | APHA, 2120 B, Visual Comparison Method           | *BDL (**DL 1.0 Hazen) | Hazen | 5                                    | 15                                                               |
| 3.     | Turbidity                           | APHA, 2130 B, Nephelometric Method               | *BDL (**DL 1.0 NTU)   | NTU   | 1                                    | 5                                                                |
| 4.     | Total Hardness as CaCO <sub>3</sub> | APHA, 2340 C, EDTA Titrimetric Method            | 171.00                | mg/l  | 200                                  | 600                                                              |
| 5.     | Calcium as Ca                       | APHA, 3500 Ca B, EDTA Titrimetric Method         | 42.53                 | mg/l  | 75                                   | 200                                                              |
| 6.     | Alkalinity as CaCO <sub>3</sub>     | APHA, 2320 B, Titration Method                   | 137.62                | mg/l  | 200                                  | 600                                                              |
| 7.     | Chloride as Cl                      | APHA, 4500 ClB, Argentometric Method             | 35.46                 | mg/l  | 250                                  | 1000                                                             |
| 8.     | Magnesium as Mg                     | APHA, 3500 Mg B, Calculation Method              | 16.04                 | mg/l  | 30                                   | 100                                                              |
| 9.     | Total Suspended solids              | APHA, 2540 C, Gravimetric Method                 | 4.12                  | mg/l  | --                                   | --                                                               |
| 10.    | Sulphate as SO <sub>4</sub>         | APHA, 4500 E, Turbidimetric Method               | 12.48                 | mg/l  | 200                                  | 400                                                              |
| 11.    | Fluoride as F                       | APHA, 4500 F D Spands Method                     | 0.52                  | mg/l  | 1.0                                  | 1.5                                                              |
| 12.    | BOD (3 Days at 27°C)                | APHA, 5210 C/ IS: 3025 (Part-44)                 | *BDL (**DL-2.0 mg/l)  | mg/l  | --                                   | --                                                               |
| 13.    | COD                                 | APHA, 5220 B, Open Reflux Method                 | *BDL (**DL-4.0 mg/l)  | mg/l  | --                                   | --                                                               |
| 14.    | Nitrate as NO <sub>3</sub>          | IS:3025 (Part-34), Chromotropic Method           | 6.95                  | mg/l  | 45                                   | No Relaxation                                                    |
| 15.    | Iron as Fe                          | APHA, 3500 Fe B 1,10 Phenanthroline Method       | 0.26                  | mg/l  | 1.0                                  | No relaxation                                                    |
| 16.    | Copper as Cu                        | APHA, 3111 B, Direct Acetylene Flame Method      | 0.38                  | mg/l  | 0.05                                 | 1.5                                                              |

Remark: - Above test results confirm that this water is fit for drinking purpose as almost all the tested parameters are well within Acceptable limits.

MEENU KAUSHIK  
SR. ANALYST

(Tested By)

ARJUN RAWAT

(Checked By)



www.vardan.co.in

Ph: 0124-4343750/752/753, 9810355569, 9953147268 E-mail: lab@vardanenvironet.com, bd@vardanenvironet.com



# Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)  
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

## Test Report

|                               |                                                                                                             |                      |                          |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------|--------------------------|
| Sample Number:                | VEL/HCL/W/07                                                                                                | Report No.:          | VEL/W/2106/24/007        |
| Name & Address of the Party:  | M/s Hindustan Copper Limited,<br>Indian Copper Complex,<br>Ghatsila, Disst.- Singhbhum,<br>Jharkhand India. | Format No.:          | 7.8 F-01                 |
| Sample Description:           | Surface Water Sample                                                                                        | Party Reference No.: | NIL                      |
| Sampling Location:            | Nitra Nala (Kendadih Mines)                                                                                 | Reporting Date:      | 29/06/2021               |
| Sample Collected By:          | VEL Representative (Mr. Kamal)                                                                              | Period of Analysis:  | 24/06/2021 to 29/06/2021 |
| Sampling & Analysis Protocol: | IS 3025 & APHA, 23 <sup>rd</sup> Edition 2017                                                               | Receipt Date:        | 24/06/2021               |
| Parameter Required:           | As Per Work Order                                                                                           | Sampling Date:       | 21/06/2021               |
|                               |                                                                                                             | Sampling Quantity:   | Grab                     |
|                               |                                                                                                             | Sampling Type:       | 2.0 Ltr. + 200 ml        |
|                               |                                                                                                             | Preservation:        | Refrigerated             |

| S. No. | Parameter                           | Test Method                                      | Result               | Unit  | Surface Water Quality Standard as per IS: 2296 (Class C) |
|--------|-------------------------------------|--------------------------------------------------|----------------------|-------|----------------------------------------------------------|
| 1.     | pH (at 25°C)                        | APHA, 4500 H <sup>+</sup> B Electrometric Method | 7.59                 | --    | 6.5-8.5                                                  |
| 2.     | Colour                              | APHA, 2120 B, Visual Comparison Method           | *BDL (**DL 5Hazen)   | Hazen | 300                                                      |
| 3.     | Turbidity                           | APHA, 2130 B, Nephelometric Method               | 1.5                  | NTU   | --                                                       |
| 4.     | Total Hardness as CaCO <sub>3</sub> | APHA, 2340 C, EDTA Titrimetric Method            | 157.00               | mg/l  | --                                                       |
| 5.     | Calcium as Ca                       | APHA, 3500 Ca B, EDTA Titrimetric Method         | 41.25                | mg/l  | --                                                       |
| 6.     | Alkalinity as CaCO <sub>3</sub>     | APHA, 2320 B, Titration Method                   | 111.48               | mg/l  | --                                                       |
| 7.     | Chloride as Cl                      | APHA, 4500 ClB, Argentometric Method             | 30.47                | mg/l  | 600                                                      |
| 8.     | Magnesium as Mg                     | APHA, 3500 Mg B, Calculation Method              | 12.93                | mg/l  | --                                                       |
| 9.     | Total Suspended solids              | APHA, 2540 C, Gravimetric Method                 | 5.7                  | mg/l  | --                                                       |
| 10.    | Sulphate as SO <sub>4</sub>         | APHA, 4500 E, Turbidimetric Method               | 19.52                | mg/l  | 400                                                      |
| 11.    | Fluoride as F                       | APHA, 4500 F D Spands Method                     | 0.65                 | mg/l  | 1.5                                                      |
| 12.    | BOD (3 Days at 27°C)                | APHA, 5210 C/ IS: 3025 (Part-44)                 | *BDL (**DL 2.0 mg/l) | mg/l  | 3                                                        |
| 13.    | COD                                 | APHA, 5220 B, Open Reflux Method                 | 9.34                 | mg/l  | --                                                       |
| 14.    | Nitrate as NO <sub>3</sub>          | IS:3025 (Part-34), Chromotropic Method           | 8.47                 | mg/l  | 50                                                       |
| 15.    | Iron as Fe                          | APHA, 3500 Fe B 1,10 Phenanthroline Method       | 0.56                 | mg/l  | 50                                                       |
| 16.    | Copper as Cu                        | APHA, 3111 B, Direct Acetylene Flame Method      | 0.76                 | mg/l  | 1.5                                                      |

Remark:-All parameters are well within limits as per Surface Water Quality Standard as per IS: 2296, Class 'C', this water may use for general purpose.

MEENU KAUSHIK  
Sr. ANALYST  
(Tested By)

ARJUN RAWAT  
(Checked By)





# Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)  
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

## Test Report

Sample Number: VEL/HCL/W/08  
Name & Address of the Party: M/s Hindustan Copper Limited,  
Indian Copper Complex,  
Ghatsila, Disst.- Singhbhum,  
Jharkhand India.  
Sample Description: Surface Water Sample  
Sampling Location: Pond Near Community Hospital  
(Kendadih Mines)  
Sample Collected By: VEL Representative (Mr. Kamal)  
Sampling & Analysis Protocol: IS 3025 & APHA, 23<sup>rd</sup> Edition 2017  
Parameter Required: As Per Work Order

Report No.: VEL/W/2106/24/008  
Format No.: 7.8 F-01  
Party Reference No.: NIL  
Reporting Date: 29/06/2021  
Period of Analysis: 24/06/2021 to 29/06/2021  
Receipt Date: 24/06/2021  
Sampling Date: 21/06/2021  
Sampling Quantity: Grab  
Sampling Type: 2.0 Ltr. + 200 ml  
Preservation: Refrigerated

| S. No. | Parameter                           | Test Method                                      | Result               | Unit  | Surface Water<br>Quality Standard<br>as per IS: 2296<br>(Class C) |
|--------|-------------------------------------|--------------------------------------------------|----------------------|-------|-------------------------------------------------------------------|
| 1.     | pH (at 25°C)                        | APHA, 4500 H <sup>+</sup> B Electrometric Method | 7.33                 | --    | 6.5-8.5                                                           |
| 2.     | Colour                              | APHA, 2120 B, Visual Comparison Method           | *BDL (**DL 5Hazen)   | Hazen | 300                                                               |
| 3.     | Turbidity                           | APHA, 2130 B, Nephelometric Method               | 2.2                  | NTU   | --                                                                |
| 4.     | Total Hardness as CaCO <sub>3</sub> | APHA, 2340 C, EDTA Titrimetric Method            | 125.14               | mg/l  | --                                                                |
| 5.     | Calcium as Ca                       | APHA, 3500 Ca B, EDTA Titrimetric Method         | 37.86                | mg/l  | --                                                                |
| 6.     | Alkalinity as CaCO <sub>3</sub>     | APHA, 2320 B, Titration Method                   | 101.24               | mg/l  | --                                                                |
| 7.     | Chloride as Cl                      | APHA, 4500 ClB, Argentometric Method             | 25.46                | mg/l  | 600                                                               |
| 8.     | Magnesium as Mg                     | APHA, 3500 Mg B, Calculation Method              | 7.19                 | mg/l  | --                                                                |
| 9.     | Total Suspended solids              | APHA, 2540 C, Gravimetric Method                 | 7.6                  | mg/l  | --                                                                |
| 10.    | Sulphate as SO <sub>4</sub>         | APHA, 4500 E, Turbidimetric Method               | 20.51                | mg/l  | 400                                                               |
| 11.    | Fluoride as F                       | APHA, 4500 F D Spands Method                     | 0.51                 | mg/l  | 1.5                                                               |
| 12.    | BOD (3 Days at 27°C)                | APHA, 5210 C/ IS: 3025 (Part-44)                 | *BDL (**DL 2.0 mg/l) | mg/l  | 3                                                                 |
| 13.    | COD                                 | APHA, 5220 B, Open Reflux Method                 | 14.63                | mg/l  | --                                                                |
| 14.    | Nitrate as NO <sub>3</sub>          | IS:3025 (Part-34), Chromotropic Method           | 12.42                | mg/l  | 50                                                                |
| 15.    | Iron as Fe                          | APHA, 3500 Fe B 1,10 Phenanthroline Method       | 0.23                 | mg/l  | 50                                                                |
| 16.    | Copper as Cu                        | APHA, 3111 B, Direct Acetylene Flame<br>Method   | 0.46                 | mg/l  | 1.5                                                               |

Remark:-All parameters are well within limits as per Surface Water Quality Standard as per IS: 2296, Class 'C', this water may use for general purpose.

MEENU KAUSHIK

(Electronically)

ANALYST

ARJUN RAWAT

(Checked By)

PS



www.vardan.co.in

Ph: 0124-4343750/752/753, 9810355569, 9953147268 E-mail: lab@vardanenvironet.com, bd@vardanenvironet.com



# Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)  
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

## Test Report

**Sample Number:** VEL/HCL/W/09  
**Name & Address of the Party:** M/s Hindustan Copper Limited,  
Indian Copper Complex,  
Ghatsila, Disst.- Singhbhum,  
Jharkhand India.  
**Sample Description:** Recycle Water Sample  
**Sampling Location:** Recycle Water Sample No.- 1  
(Kendadih Mines)  
**Sample Collected By:** VEL Representative (Mr. Kamal)  
**Sampling & Analysis Protocol:** IS: 10500-2012 & APHA  
**Parameter Required:** As Per Work Order  
**Report No.:** VEL/W/2106/24/009  
**Format No.:** 7.8 F-01  
**Party Reference No.:** NIL  
**Reporting Date:** 29/06/2021  
**Period of Analysis:** 24/06/2021 to 29/06/2021  
**Receipt Date:** 24/06/2021  
**Sampling Date:** 21/06/2021  
**Sampling Quantity:** Grab  
**Sampling Type:** 2.0 Ltr. + 200 ml  
**Preservation:** Refrigerated

| S. No. | Parameter                           | Test Method                                      | Result                | Unit  |
|--------|-------------------------------------|--------------------------------------------------|-----------------------|-------|
| 1.     | pH (at 25 C)                        | APHA, 4500 H <sup>+</sup> B Electrometric Method | 6.84                  | --    |
| 2.     | Colour                              | APHA, 2120 B, Visual Comparison Method           | *BDL (**DL 5 Hazen)   | Hazen |
| 3.     | Turbidity                           | APHA, 2130 B, Nephelometric Method               | 1.5                   | NTU   |
| 4.     | Total Hardness as CaCO <sub>3</sub> | APHA, 2340 C, EDTA Titrimetric Method            | 154.60                | mg/l  |
| 5.     | Calcium as Ca                       | APHA, 3500 Ca B, EDTA Titrimetric Method         | 42.84                 | mg/l  |
| 6.     | Alkalinity as CaCO <sub>3</sub>     | APHA, 2320 B, Titration Method                   | 135.26                | mg/l  |
| 7.     | Chloride as Cl                      | APHA, 4500 Cl B, Argentometric Method            | 34.11                 | mg/l  |
| 8.     | Magnesium as Mg                     | APHA, 3500 Mg B, Calculation Method              | 11.96                 | mg/l  |
| 9.     | Total Suspended solids              | APHA, 2540 C, Gravimetric Method                 | <5.0                  | mg/l  |
| 10.    | Sulphate as SO <sub>4</sub>         | APHA, 4500 E, Turbidimetric Method               | 10.65                 | mg/l  |
| 11.    | Fluoride as F                       | APHA, 4500 F D Spands Method                     | 0.81                  | mg/l  |
| 12.    | BOD (3 Days at 27°C)                | APHA, 5210 C/ IS: 3025 (Part-44)                 | <5.0                  | mg/l  |
| 13.    | COD                                 | APHA, 5220 B, Open Reflux Method                 | <5.0                  | mg/l  |
| 14.    | Nitrate as NO <sub>3</sub>          | IS:3025 (Part-34), Chromotropic Method           | 0.79                  | mg/l  |
| 15.    | Iron as Fe                          | APHA, 3500 Fe B 1,10 Phenanthroline Method       | 0.23                  | mg/l  |
| 16.    | Copper as Cu                        | APHA, 3111 B, Direct Acetylene Flame Method      | 0.26                  | mg/l  |
| 17.    | Lead as Pb                          | APHA, 3111 B, Direct Acetylene Flame Method      | *BDL (**DL 0.01 mg/l) | mg/l  |
| 18.    | Oil & Grease                        | APHA, 5520 B, Partition Gravimetric Method       | <1.0                  | mg/l  |

Remark:- This water is suitable for non potable purpose. \*BDL- Below Detection Limit, \*\*DL- Detection Limit

MEENU KAUSHIK  
SR. ANALYST  
(Tested By)

ARJUN RAWAT  
(Checked By)



www.vardan.co.in

Ph: 0124-4343750/752/753, 9810355569, 9953147268 E-mail: lab@vardanenvironet.com, bd@vardanenvironet.com



# Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)  
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

## Test Report

|                               |                                                                                                             |                      |                          |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------|--------------------------|
| Sample Number:                | VEL/HCL/W/10                                                                                                | Report No.:          | VEL/W/2106/24/010        |
| Name & Address of the Party:  | M/s Hindustan Copper Limited,<br>Indian Copper Complex,<br>Ghatsila, Distt.- Singhbhum,<br>Jharkhand India. | Format No.:          | 7.8 F-01                 |
| Sample Description:           | Recycle Water Sample                                                                                        | Party Reference No.: | NIL                      |
| Sampling Location:            | Recycle Water Sample No.- 2<br>(Kendadih Mines)                                                             | Reporting Date:      | 29/06/2021               |
| Sample Collected By:          | VEL Representative (Mr. Kamal)                                                                              | Period of Analysis:  | 24/06/2021 to 29/06/2021 |
| Sampling & Analysis Protocol: | IS: 10500-2012 & APHA                                                                                       | Receipt Date:        | 24/06/2021               |
| Parameter Required:           | As Per Work Order                                                                                           | Sampling Date:       | 21/06/2021               |
|                               |                                                                                                             | Sampling Quantity:   | Grab                     |
|                               |                                                                                                             | Sampling Type:       | 2.0 Ltr. + 200 ml        |
|                               |                                                                                                             | Preservation:        | Refrigerated             |

| S. No. | Parameter                           | Test Method                                      | Result                | Unit  |
|--------|-------------------------------------|--------------------------------------------------|-----------------------|-------|
| 1.     | pH (at 25°C)                        | APHA, 4500 H <sup>+</sup> B Electrometric Method | 7.52                  | --    |
| 2.     | Colour                              | APHA, 2120 B, Visual Comparison Method           | *BDL (**DL 5Hazen)    | Hazen |
| 3.     | Turbidity                           | APHA, 2130 B, Nephelometric Method               | 1.6                   | NTU   |
| 4.     | Total Hardness as CaCO <sub>3</sub> | APHA, 2340 C, EDTA Titrimetric Method            | 286.24                | mg/l  |
| 5.     | Calcium as Ca                       | APHA, 3500 Ca B, EDTA Titrimetric Method         | 99.52                 | mg/l  |
| 6.     | Alkalinity as CaCO <sub>3</sub>     | APHA, 2320 B, Titration Method                   | 296.23                | mg/l  |
| 7.     | Chloride as Cl                      | APHA, 4500 Cl B, Argentometric Method            | 85.14                 | mg/l  |
| 8.     | Magnesium as Mg                     | APHA, 3500 Mg B, Calculation Method              | 9.62                  | mg/l  |
| 9.     | Total Suspended solids              | APHA, 2540 C, Gravimetric Method                 | <5.0                  | mg/l  |
| 10.    | Sulphate as SO <sub>4</sub>         | APHA, 4500 E, Turbidimetric Method               | 22.45                 | mg/l  |
| 11.    | Fluoride as F                       | APHA, 4500 F D Spands Method                     | 0.76                  | mg/l  |
| 12.    | BOD (3 Days at 27°C)                | APHA, 5210 C/ IS: 3025 (Part-44)                 | <5.0                  | mg/l  |
| 13.    | COD                                 | APHA, 5220 B, Open Reflux Method                 | <5.0                  | mg/l  |
| 14.    | Nitrate as NO <sub>3</sub>          | IS:3025 (Part-34). Chromotropic Method           | 0.81                  | mg/l  |
| 15.    | Iron as Fe                          | APHA, 3500 Fe B 1, 10 Phenanthroline Method      | 0.21                  | mg/l  |
| 16.    | Copper as Cu                        | APHA, 3111 B, Direct Acetylene Flame Method      | 0.15                  | mg/l  |
| 17.    | Lead as Pb                          | APHA, 3111 B, Direct Acetylene Flame Method      | *BDL (**DL 0.01 mg/l) | mg/l  |
| 18.    | Oil & Grease                        | APHA, 5520 B, Partition Gravimetric Method       | <1.0                  | mg/l  |

Remark:- This water is suitable for non potable purpose. \*BDL- Below Detection Limit, \*\*DL- Detection Limit

MEENU KAUSHIK  
SR ANALYST

(Tested By)

ARJUN RAWAT

(Checked By)



VEL/E/N/TR/PN1369

[www.vardan.co.in](http://www.vardan.co.in)

Ph: 0124-4343750/752/753, 9810355569, 9953147268 E-mail: [lab@vardanenviromet.com](mailto:lab@vardanenviromet.com), [bd@vardanenviromet.com](mailto:bd@vardanenviromet.com)



# Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)  
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

## Test Report

|                               |                                                                                                             |                      |                          |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------|--------------------------|
| Sample Number:                | VEL/HCL/W/09                                                                                                | Report No.:          | VEL/W/2104/26/009        |
| Name & Address of the Party:  | M/s Hindustan Copper Limited,<br>Indian Copper Complex,<br>Ghatsila, Disst.- Singhbhum,<br>Jharkhand India. | Format No.:          | 7.8 F-01                 |
| Sample Description:           | Recycle Water Sample                                                                                        | Party Reference No.: | NIL                      |
| Sampling Location:            | Recycle Water Sample No.- 1<br>(Kendadih Mines)                                                             | Reporting Date:      | 03/05/2021               |
| Sample Collected By:          | VEL Representative (Mr. Kamal)                                                                              | Period of Analysis:  | 26/04/2021 to 03/05/2021 |
| Sampling & Analysis Protocol: | IS: 10500-2012 & APHA                                                                                       | Receipt Date:        | 26/04/2021               |
| Parameter Required:           | As Per Work Order                                                                                           | Sampling Date:       | 22/04/2021               |
|                               |                                                                                                             | Sampling Quantity:   | 2.0 Ltr.                 |
|                               |                                                                                                             | Sampling Type:       | Grab                     |
|                               |                                                                                                             | Preservation:        | Refrigerated             |

| S. No. | Parameter                           | Test Method                                      | Result                | Unit  |
|--------|-------------------------------------|--------------------------------------------------|-----------------------|-------|
| 1.     | pH (at 25°C)                        | APHA, 4500 H <sup>+</sup> B Electrometric Method | 6.84                  | --    |
| 2.     | Colour                              | APHA, 2120 B, Visual Comparison Method           | *BDL (**DL 5 Hazen)   | Hazen |
| 3.     | Turbidity                           | APHA, 2130 B, Nephelometric Method               | 1.5                   | NTU   |
| 4.     | Total Hardness as CaCO <sub>3</sub> | APHA, 2340 C, EDTA Titrimetric Method            | 154.60                | mg/l  |
| 5.     | Calcium as Ca                       | APHA, 3500 Ca B, EDTA Titrimetric Method         | 42.84                 | mg/l  |
| 6.     | Alkalinity as CaCO <sub>3</sub>     | APHA, 2320 B, Titration Method                   | 135.26                | mg/l  |
| 7.     | Chloride as Cl                      | APHA, 4500 Cl B, Argentometric Method            | 34.11                 | mg/l  |
| 8.     | Magnesium as Mg                     | APHA, 3500 Mg B, Calculation Method              | 11.59                 | mg/l  |
| 9.     | Total Suspended solids              | APHA, 2540 C, Gravimetric Method                 | <5.0                  | mg/l  |
| 10.    | Sulphate as SO <sub>4</sub>         | APHA, 4500 E, Turbidimetric Method               | 10.65                 | mg/l  |
| 11.    | Fluoride as F                       | APHA, 4500 F D Spands Method                     | 0.81                  | mg/l  |
| 12.    | BOD (3 Days at 27°C)                | APHA, 5210 C/ IS: 3025 (Part-44)                 | <5.0                  | mg/l  |
| 13.    | COD                                 | APHA, 5220 B, Open Reflux Method                 | <5.0                  | mg/l  |
| 14.    | Nitrate as NO <sub>3</sub>          | IS:3025 (Part-34), Chromotropic Method           | 0.79                  | mg/l  |
| 15.    | Iron as Fe                          | APHA, 3500 Fe B 1,10 Phenanthroline Method       | 0.23                  | mg/l  |
| 16.    | Copper as Cu                        | APHA, 3111 B, Direct Acetylene Flame Method      | 0.26                  | mg/l  |
| 17.    | Lead as Pb                          | APHA, 3111 B, Direct Acetylene Flame Method      | *BDL (**DL 0.01 mg/l) | mg/l  |
| 18.    | Oil & Grease                        | APHA, 5520 B, Partition Gravimetric Method       | <1.0                  | mg/l  |

Remark:- This water is suitable for non potable purpose. \*BDL- Below Detection Limit, \*\*DL- Detection Limit

RANJAN SHARMA  
Jr. (Tested By)

ARJUN RAWAT  
(Checked By)



VEL/E/N/TR/PN1824

www.vardan.co.in



# Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)  
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

## Test Report

|                               |                                                                                                             |                      |                          |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------|--------------------------|
| Sample Number:                | VEL/HCL/W/10                                                                                                | Report No.:          | VEL/W/2104/26/010        |
| Name & Address of the Party:  | M/s Hindustan Copper Limited,<br>Indian Copper Complex,<br>Ghatsila, Disst.- Singhbhum,<br>Jharkhand India. | Format No.:          | 7.8 F-01                 |
| Sample Description:           | Recycle Water Sample                                                                                        | Party Reference No.: | NIL                      |
| Sampling Location:            | Recycle Water Sample No.- 2<br>(Kendadih Mines)                                                             | Reporting Date:      | 03/05/2021               |
| Sample Collected By:          | VEL Representative (Mr. Kamal)                                                                              | Period of Analysis:  | 26/04/2021 to 03/05/2021 |
| Sampling & Analysis Protocol: | IS: 10500-2012 & APHA                                                                                       | Receipt Date:        | 26/04/2021               |
| Parameter Required:           | As Per Work Order                                                                                           | Sampling Date:       | 22/04/2021               |
|                               |                                                                                                             | Sampling Quantity:   | 2.0 Ltr.                 |
|                               |                                                                                                             | Sampling Type:       | Grab                     |
|                               |                                                                                                             | Preservation:        | Refrigerated             |

| S. No. | Parameter                           | Test Method                                      | Result                | Unit  |
|--------|-------------------------------------|--------------------------------------------------|-----------------------|-------|
| 1.     | pH (at 25°C)                        | APHA, 4500 H <sup>+</sup> B Electrometric Method | 7.38                  | --    |
| 2.     | Colour                              | APHA, 2120 B, Visual Comparison Method           | *BDL (**DL 5Hazen)    | Hazen |
| 3.     | Turbidity                           | APHA, 2130 B, Nephelometric Method               | 1.8                   | NTU   |
| 4.     | Total Hardness as CaCO <sub>3</sub> | APHA, 2340 C, EDTA Titrimetric Method            | 295.30                | mg/l  |
| 5.     | Calcium as Ca                       | APHA, 3500 Ca B, EDTA Titrimetric Method         | 102.32                | mg/l  |
| 6.     | Alkalinity as CaCO <sub>3</sub>     | APHA, 2320 B, Titration Method                   | 301.84                | mg/l  |
| 7.     | Chloride as Cl                      | APHA, 4500 Cl B, Argentometric Method            | 89.22                 | mg/l  |
| 8.     | Magnesium as Mg                     | APHA, 3500 Mg B, Calculation Method              | 9.72                  | mg/l  |
| 9.     | Total Suspended solids              | APHA, 2540 C, Gravimetric Method                 | <5.0                  | mg/l  |
| 10.    | Sulphate as SO <sub>4</sub>         | APHA, 4500 E, Turbidimetric Method               | 25.71                 | mg/l  |
| 11.    | Fluoride as F                       | APHA, 4500 F D Spands Method                     | 0.82                  | mg/l  |
| 12.    | BOD (3 Days at 27°C)                | APHA, 5210 C/ IS: 3025 (Part-44)                 | <5.0                  | mg/l  |
| 13.    | COD                                 | APHA, 5220 B, Open Reflux Method                 | <5.0                  | mg/l  |
| 14.    | Nitrate as NO <sub>3</sub>          | IS:3025 (Part-34), Chromotropic Method           | 0.88                  | mg/l  |
| 15.    | Iron as Fe                          | APHA, 3500 Fe B 1,10 Phenanthroline Method       | 0.24                  | mg/l  |
| 16.    | Copper as Cu                        | APHA, 3111 B, Direct Acetylene Flame Method      | 0.19                  | mg/l  |
| 17.    | Lead as Pb                          | APHA, 3111 B, Direct Acetylene Flame Method      | *BDL (**DL 0.01 mg/l) | mg/l  |
| 18.    | Oil & Grease                        | APHA, 5520 B, Partition Gravimetric Method       | <1.0                  | mg/l  |

Remark:- This water is suitable for non potable purpose. \*BDL- Below Detection Limit, \*\*DL- Detection Limit

KANCHAN SHARMA  
(Tested By)  
Jr. Lab Analyst

ARJUN RAWAT  
(Checked By)  
Sr. Lab Analyst



VEL/E/N/TR/PN1823

www.vardan.co.in

Ph: 0124-4343750/752/753, 9810355569, 9953147268 E-mail: lab@vardanenvironet.com, bd@vardanenvironet.com